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Ophelia: Week 5, Day 5 - 10 RAG Advanced Techniques

10 RAG Advanced Techniques

1. Chunking R&D: experiment with chunking strategy
2. Encoder R&D: select the best Encoder model based on a test set
3. Improve prompts: General Context, the current date, relevant context and history
4. Document pre-processing: use an LLM to make the chunks and/or text for encoding.
5. Query rewriting: use an LLM to convert the user's question to a RAG query.
6. Query expansion: use an LLM to turn the question into multiple RAG queries
7. Re-ranking: use an LLM to sub-select from RAG results.
8. Hierarchical: use an LLM to summarize at multiple levels
9. Graph RAG: retrieve content closely related to similar documents
10. Agentic RAG: use Agents for retrieval, combining with memory and Tools such as SQL

Steps in doing LLM Semantic Rewriting

1. Fetch documents from the knowledge base, like LangChain
2. Call an LLM to turn documents into chunks
3. Store the chunks in Chroma